Ritesh Somashekar

AI & ML Engineer

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PROFESSIONAL SUMMARY

- **AI & Machine Learning Engineer** with **5+ years** of experience progressing from **data engineering** to **machine learning** and **Generative AI**, building **production RAG systems**, **LLM applications**, and **enterprise data pipelines** across healthcare and public safety. Published researcher (**ACM WSDM '26**, **IEEE PST 2026**) with hands-on expertise in **Llama 3, LlamaIndex, ChromaDB, pgvector, FAISS, Hugging Face, and PyTorch**.
- Strong background in **data engineering** with **Apache Airflow, Spark/PySpark, Snowflake, and Databricks**, and **cloud pipelines on AWS (S3, Lambda, Athena, Glue, SageMaker)** and **Azure**, with a focus on **HIPAA-compliant** healthcare data. Experienced in **MLOps** using **MLflow, Docker, Kubernetes, FastAPI,** and **CI/CD pipelines**.
- Demonstrated impact, including **RAG recall@10 raised from 61% to 88%**, a **0.81 protocol F1 with a 3.0% hallucination rate**, **99.9% pipeline reliability**, and **60% faster model training**, delivering **end-to-end AI solutions** from research to production.

TECHNICAL SKILLS

Programming & Scripting: Python (NumPy, pandas, scikit-learn, PyTorch, TensorFlow), SQL, R, Scala, Java, Bash

RAG & Generative AI: Retrieval-Augmented Generation (RAG), Chunking, Embeddings (Sentence Transformers, all-mpnet-base-v2), Metadata Filtering, Top-K Retrieval, Context Assembly, Grounding, LlamaIndex, Vector Databases (ChromaDB, pgvector, FAISS), Llama 3/3.1, Ollama, Prompt Engineering, Agentic AI, LoRA/PEFT, Quantization-Aware Training (QAT)

LLM Evaluation & Safety: RAGAS, Recall@K, Faithfulness, Hallucination Detection, Guardrails, Ontology-Based Validation, NLI-Based Validation, Semantic Similarity Scoring, Ablation Studies, Protocol F1, Confidence Scoring

Machine Learning & NLP: Supervised & Unsupervised Learning, Transformers, BERT, BioBERT, Hugging Face, XGBoost, LightGBM, Graph Analytics, Ensemble Models, Time Series Forecasting, Anomaly Detection, Feature Engineering, Model Explainability (SHAP)

MLOps & ML Infrastructure: MLflow, Docker, Kubernetes, Jenkins, CI/CD, FastAPI, Git, DVC, Model Deployment

Data Engineering & Big Data: Apache Airflow, Apache Spark, PySpark, Kafka, Databricks, Snowflake, Delta Lake, Hadoop, Hive, Sqoop, Informatica PowerCenter, Precisely Data360, SQL Server, MySQL, MongoDB

Cloud & Visualization: AWS (S3, Lambda, Athena, Glue, SageMaker, SQS), Azure (Data Services, Stream Analytics), Tableau, Power BI (DAX), Excel (VBA)

EDUCATION

Master's in Data Analytics and Engineering | George Mason University, Fairfax, VA, USA | GPA: 3.93/4.0

Bachelor's in Information Science and Engineering | Visvesvaraya Technological University, Bengaluru, Karnataka, India.

PROFESSIONAL EXPERIENCE

George Mason University | ***AI & ML Research Associate*** **Fairfax, VA | August 2024 - Present**

- Co-developed **SHIELD** (published at **ACM WSDM '26**), an online conversational AI training platform for 9-1-1 call-takers built with **Fairfax County DPSC**, using **LLM agents** to generate call scenarios from CAD data, deliver **real-time metacognitive nudges**, and produce post-call reflection analytics.
- Co-developed **MATE-911** (published at **IEEE PST 2026**), a multimodal **RAG** system that grounds **Llama 3.1 8B** generation in validated emergency-response knowledge, producing protocol-compliant voice and text scenarios on commodity hardware with **0.81 protocol F1**, a **3.0% hallucination rate**, and a **97.6% pass rate**, with no commercial API dependency.
- **Led the RAG knowledge-base and data pipeline for both platforms** in **Python**, covering OCR parsing, abbreviation expansion, completeness checks, segmentation, and standardization of **160 gold-standard scenarios**; the curated knowledge base cut hallucinations from **13.0% to 3.0%**.
- Built the **retrieval layer** with **all-mpnet-base-v2** (768-dim) embeddings in a **ChromaDB HNSW cosine index**, attaching metadata (category, severity, tags, confidence) for **filtered top-K retrieval** and a **70% similarity relevance gate**.
- Engineered **prompt and context assembly** to ground **Llama 3.1 8B (Ollama)** responses in retrieved scenarios, tuning temperature, token budgets, and context windows to reach **960 ms (text)** and **1.3 s (voice)** mean latency, with fallback generation for resilience.

- **Led the validation layer for both platforms**, building **RAG guardrails** with a **911 ontology validator** (NENA standards, DPSC procedures), rule-based and **LLM-based semantic checks**, **NLI-based hallucination detection**, PASS/REVIEW/FAIL confidence scoring, and an **85% cosine-similarity** quality gate.
- Ran **RAG ablation studies** showing retrieval drives quality (**F1 0.81 vs. 0.59** and **97.6% vs. 49.7%** pass rate without it), while ontology validation acts as a safety guardrail that cuts hallucinations from **9.5% to 3.0%**.
- Developed a **RAG research assistant** with **LlamaIndex**, **pgvector**, and **FAISS**, optimizing chunking, embeddings, retrieval, and context assembly to raise **recall@10 from 61% to 88%**, evaluated with **RAGAS** (faithfulness, answer relevance, context precision) and served to **40+ researchers** at about **190 ms p95** latency.
- Applied **graph-based analytics** and **ensemble models** to large-scale disaster datasets on the **ORC Hopper HPC cluster**, forecasting community coping trends with an **RMSE of 0.89**.
- Implemented **model explainability** using **SHAP** for **intent classification** and **behavioral anomaly detection** models, guiding threshold tuning and model governance; the **LightGBM** anomaly detector reduced false positives by **25%**.
- Fine-tuned **Llama 3** with **LoRA/PEFT** and **QAT**, cutting model size by **40%** within **3%** of full-precision accuracy.
- Took models and RAG services from **proof of concept to production** with **MLflow** experiment tracking and model registry, **FastAPI**, **Docker**, and **CI/CD**, reducing release cycles from **2 weeks to 5 days** and enabling **A/B testing** of model and prompt versions.

Carelon Global Solutions (Elevance Health) | ***Machine Learning Engineer*** **Bengaluru, India | May 2022 - August 2023**

- Built NLP models using **BERT** and **BioBERT** for clinical entity extraction from unstructured medical text, achieving **80% accuracy** and reducing processing time by **55%**.
- Improved **Scikit-learn** and **XGBoost** models through feature engineering and hyperparameter tuning, increasing accuracy by **8%** and cutting training time by **15%**.
- Containerized ML inference services with **Docker, Kubernetes,** and **Jenkins CI/CD**, improving system uptime by **20%** and enabling real-time EHR integration via **REST APIs**.
- Engineered ML-ready ingestion of **400K+ XML/JSON healthcare records** into **Snowflake** and **Databricks**, accelerating model training by **60%** and reducing preprocessing time by **70%**.
- Built a daily **NPPES** ingestion workflow that improved ML data accuracy by **74%** and cut processing overhead by **60%**, and built **PySpark** ETL on **AWS (Glue, SageMaker)** with **Delta Lake**, improving data reliability by **30%** under **HIPAA**.
- Designed **20+ enterprise ETL pipelines** orchestrated with **Apache Airflow** and **AWS (S3, Lambda, Athena)**, boosting throughput by **70%** and achieving **99.9% pipeline reliability** with end-to-end lineage tracking.

Carelon Global Solutions (Elevance Health) | ***Data Engineer*** **Bengaluru, India | October 2020 - April 2022**

- Aligned ETL pipelines with business KPIs on the **Seven Plus Locations** project, improving claims data accuracy.
- Redesigned **claims-balancing ETL workflows** and automated **medical invoice processing** pipelines, reducing operational overhead by **60%** and manual errors by **70%**.
- Migrated large-scale processing workloads from **Hive** to **Apache Spark/PySpark**, and moved **Sqoop/Hadoop** ingestion to **Azure Data Services**, improving performance, scalability, and pipeline efficiency.
- Developed **RESTful microservices** with **OAuth 2.0** and API versioning, and designed **MongoDB** data models with **Python/Java** validation scripts (UDFs) to improve data quality and consistency.

PROJECTS

Agentic AI-Based Ticket Automation System | SecureGPT, LLaMA, AWS SQS, JIRA, NIST-CSF

- Built an **agentic AI** pipeline with **SecureGPT** and **LLaMA** on **AWS SQS** event queues that monitors cybersecurity threats and auto-generates **JIRA** tickets under **NIST-CSF**, reaching **93% automation accuracy** and cutting triage time by **70%**.
- Used **LLM prompt engineering** to classify and rank events, improving scalability by **50%** and cutting operational risk by **40%**.

PUBLICATIONS

- R. S. Nayaka, **R. Somashekar**, K. B. Laskey, L. Wells II, and H. Purohit. "SHIELD: A Framework for Online Conversational AI System for Self-Regulated Learning at Emergency Communications Centers." In *Proceedings of the 19th ACM International Conference on Web Search and Data Mining (WSDM '26)*, Boise, ID, 2026. doi:10.1145/3773966.3779411
- R. S. Nayaka, **R. Somashekar**, and H. Purohit. "MATE-911: Multimodal AI-powered Training Environment for Next-Generation 911 Dispatcher Preparedness." In *IEEE PST Conference*, 2026.